

Online Algebra, Logic, and Topology Seminar

Category Theory Seminar — Abstract

December 02nd, 2025

Categorical and metric density

Speaker: Tome Leinster

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Date & Time: December 02nd, 2025 (Tuesday) — 3:00 PM (Coimbra time)

Place: Google Meet

Abstract. When metric spaces are viewed as enriched categories, the categorical notion of density does not coincide with the ordinary topological notion. This prompts two questions. First, what does categorical density mean for metric spaces? And second, does the condition of topological density generalize from metric spaces to arbitrary enriched categories? Both questions were considered in Lawvere's 1973 metric spaces paper, but only very briefly: there is a lot more to say. Some satisfying new answers have recently been uncovered by my student Adrián Doña Mateo, which I will explain.